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MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTROTECHNOLOGY
NQF LEVEL 2

17 February 2026

This marking guideline consists of 7 pages.

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QUESTION 1

- 1.1 1.1.1 Molecule
 1.1.2 Diode
 1.1.3 Faraday's law of induction
 1.1.4 Element
 1.1.5 Compound
(5 × 1) (5)
- 1.2 1.2.1 The resistors are connected in a series-parallel circuit. (1)
- 1.2.2 $R_P = \frac{R_2 \times R_3}{R_2 + R_3}$
 $\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_3}$
 $\frac{1}{R_P} = \frac{1}{4} + \frac{1}{6}$
 $\frac{1}{R_P} = \frac{5}{12}$
 $\therefore R_P = 2,4 \Omega$

 $R_T = R_1 + R_P + R_4$
 $R_T = 2 + 2,4 + 3$
 $R_T = 7,4 \Omega$ ✓ (3)
- 1.2.3 $I_T = \frac{V_T}{R_T}$
 $I_T = \frac{12}{7,4}$ ✓
 $I_T = 1,622 A$ ✓ (2)
- 1.2.4 $P = V_T \times I_T$
 $P = 12 \times 1,622$ ✓
 $P = 19,464 W$ ✓ (2)
- 1.3 • Type of material of the conductor
 • Cross-sectional area of the conductor
 • Length of the conductor
 • Temperature (Any 2 × 1) (2)
- 1.4 1.4.1 $\therefore \frac{N_1}{N_2} = \frac{V_1}{V_2}$
 $V_1 = \frac{V_2 \times N_1}{N_2}$
 $V_1 = \frac{300 \times 24}{60}$ ✓✓
 $V_1 = 120 V$ ✓ (3)

1.4.2

$$\therefore \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$I_2 = \frac{N_1 \times I_1}{N_2}$$

$$I_2 = \frac{300 \times 2}{60} \checkmark$$

$$I_2 = 10 \text{ A} \checkmark$$

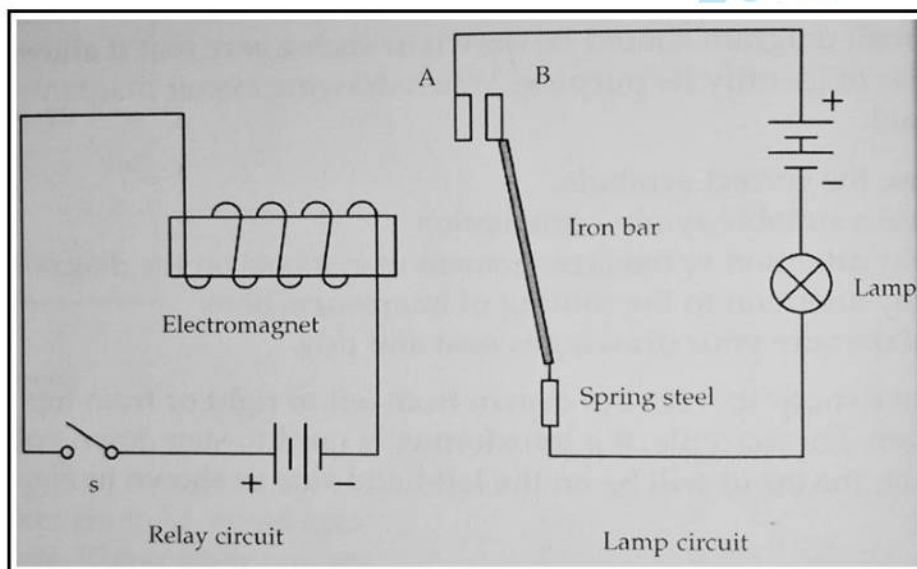
(2)
[20]

QUESTION 2

- 2.1
- Direct contact
 - Indirect contact

(2)

2.2



(6)

- 2.3
- 2.3.1 2,2 m
- 2.3.2 2,2 m
- 2.3.3 1,2 m
- 2.3.4 0,2 m

(4 × 1) (4)

- 2.4
- Less expensive than metal conduit
 - Weighs less than a metal pipe
 - Quick and easy to install
 - Easy to transport
 - Has an attractive finish
 - Has good insulating properties to prevent earth failure short circuit
 - Does not rust (is anti-corrosive)

(Any 4 × 1) (4)

- 2.5
- Screwdrivers
 - Voltage tester/Multimeter
 - Wire strippers
 - Crimping tool
 - Pliers (combination, long-nose, side-cutters)
 - Insulation tape
 - Cable test
 - Drill and drill bits
 - Measuring tape
 - Label maker or markers
- (Any 4 × 1) (4)
- 2.6 Bending spring (1)
- 2.7
- Size and type of load to be supplied
 - Permissible voltage drop
 - Prospective fault current
 - Circuit protection
 - Installation conditions
- (Any 4 × 1) (4)
- [25]

QUESTION 3

- 3.1
- The device must be fit for purpose.
 - Equipment and test instruments must be tested every six months and recalibrated once every year to ensure that they are in proper working order.
 - Repairs and maintenance should meet appropriate standards and the manufacturer's specifications.
 - Immediately before and after use, the device should be inspected and functional checks performed to confirm that the device is operating correctly.
 - The work practices employed during use should be in line with documented procedures. These documented procedures should address the manufacturer's instructions and recommendations.
 - The function, range and class accuracy of the device should be appropriate to both the job at hand and the general working conditions. The combination of leads and instruments used should be capable of withstanding the impulse voltages and fault current levels.
- (Any 5 × 1) (5)
- 3.2
- | | |
|-------|-------------|
| 3.2.1 | 1 kΩ ± 10% |
| 3.2.2 | 8,2 kΩ ± 5% |
| 3.2.3 | 910 kΩ ± 5% |
| 3.2.4 | 6,8 MΩ ± 5% |
- (4 × 1) (4)

- 3.3
- By what date must the task be completed?
 - When must the task start?
 - How many workers will be available?
 - What tools and equipment will be needed?
 - What materials must be ordered?
 - What documentation is needed according to the job instructions?
 - Must any requests be submitted to the supply authority, for example an application for an electricity connection?
 - What will happen in the case of unforeseen circumstances, such as heavy rain, unavailability of materials, etc.?
- (Any 6 × 1) (6)

3.4

ITEM NUMBER	DESCRIPTION	QUANTITY	
1	Resistor	3	✓
2	Potentiometer/Trimmer	1	✓
3	Diode	5	✓
4	LDR	1	✓
5	Capacitor	2	✓
6	NPN transistor	2	✓
7	Transformer	1	✓

(7)

- 3.5
- The resistance may drift from its normal value.
 - They may be completely open.
 - They may be mechanically damaged.
- (3)
[25]

QUESTION 4

- 4.1
- 4.1.1 True
- 4.1.2 False
- 4.1.3 False
- 4.1.4 True
- 4.1.5 True
- (5 × 1) (5)
- 4.2
- Visual inspection
 - Half-cut method
 - Input-to-output technique
 - Operational method
 - Circuit continuity test
- (5)
- 4.3
- Electric shock is a dangerous physical effect as a result of the passing of an electric current through the body of a person or animal.
- (1)

4.4 4.4.1 $R_S = R_1 + R_2$
 $R_S = 50 + 100$
 $R_S = 150 \Omega \checkmark$

$$R_P = \frac{R_S \times R_3}{R_S + R_3}$$

$$R_P = \frac{150 \times 150}{150 + 150} \checkmark$$

$$R_P = 75 \Omega \checkmark \quad (3)$$

4.4.2 $I_T = \frac{V_T}{R_T}$
 $I_T = \frac{3}{75} \checkmark$
 $I_T = 0,04 A \checkmark$ or 40 mA (2)

4.4.3 When R_3 is open:
 $R_T = R_1 + R_2 = 50 + 100 = 150 \Omega \checkmark$
 $I_T = \frac{V_T}{R_T} = \frac{3}{150} = 0,02 \Omega \checkmark$ (2)

4.4.4 When resistor (R_1) is short-circuited:
 $R_P = \frac{R_2 \times R_3}{R_2 + R_3}$
 $R_P = \frac{100 \times 150}{100 + 150}$
 $R_P = 60 \Omega \checkmark$
 $I_A = \frac{V_T}{R_T} = \frac{3}{60} = 0,05 \Omega \checkmark$ (2)

[20]

QUESTION 5

- 5.1 Step 1: Fill in the form.
Step 2: Give the form to the attending doctor.
Step 3: Send the forms to the compensation fund.
Step 4: Pay compensation to the worker.
Step 5: Send follow-up forms. (5)

- 5.2
- Decide on a goal or goals.
 - List the key tasks required to accomplish the goals.
 - Describe measurable results or outcomes for each of the key tasks.
 - Test the goals against the tasks and the expected results.
 - Arrange the tasks in the order in which they must be performed.
 - Determine the resources, time, money and manpower needed to carry out the tasks.
 - Put everything in a timeframe, indicating dates for the completion of the task. (Any 5 × 1) (5)

[10]

TOTAL: 100